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Safe Reinforcement Learning Control of High-Speed Maglev Train Levitation System Considering Aerodynamic Lift Force

Xiaoning Zhao, Yougang Sun*, Zhao Xu, Zeng Zhang, Bing Ren

***Abstract* -- High-speed maglev train levitation systems face significant stability and safety challenges due to their inherent nonlinear open-loop instability and the complex operating environment caused by aerodynamic lift force and track irregularities. Existing model-based control methods rely on precise mathematical models and manual parameter tuning, struggling to adapt to complex dynamic environments. Learning-based approaches, meanwhile, suffer from difficulties in convergence under strong disturbances and insufficient safety guarantees. To address these limitations, this paper proposes a safe reinforcement learning control method based on higher-order control barrier functions (HOCBF) and disturbance observer (DOB). The proposed method employs a hierarchical design: reinforcement learning (RL) adaptively learns the optimal policy from data; HOCBF construct a safety layer to modify the RL agent's actions, ensuring system safety; and the DOB compensates for external disturbances like aerodynamic lift, enhancing convergence stability under strong disturbances. Simulation results verify the efficacy of the presented approach under three operating conditions, confirming remarkable improvements in convergence performance under strong disturbances and enhanced stability.**

***Index Terms* -- Maglev train, Levitation system, Safe reinforcement learning, Control barrier function, Disturbance observer.**

I. Introduction

Electromagnetic Suspension (EMS) type high-speed maglev trains achieve contactless support, guidance, traction, and braking through electromagnetic forces between the vehicle and the track. Maglev trains obtain power through non-contact energy transfer, overcoming the two key limitations constraining the speed of traditional wheel-rail trains: adhesion utilization and sliding current collection [1], [2]. Consequently, maglev trains have become a research hotspot for next-generation ultra-high-speed trains. Theoretically, high-speed maglev trains offer an economically viable operating speed of up to 600 km/h, effectively bridging the speed gap between high-speed rail and air transport [3].

The levitation subsystem serves as the core component ensuring contactless and safe operation of high-speed maglev trains, and its operational performance directly impacts the train's safety and passenger ride comfort [4]. The levitation system exhibits inherent nonlinear open-loop instability. Furthermore, electromagnets can only provide unidirectional upward force, stable levitation relies on the counteracting force of gravity [5]. The operating environment is complex, subject to aerodynamic lift forces and track irregularities. At high speeds, the aerodynamic lift force increases rapidly. Its counteraction to gravity causes a rapid reduction in the levitation gap, significantly increasing the risk of collision accidents between the train and track. [6].

Numerous researchers have proposed solutions from the perspective of levitation control algorithms. For instance, considering the impact of unsteady periodic aerodynamic forces on the levitation system's dynamic response, Liu et al. [7] identified bifurcation parameter intervals for control gains in full-state feedback control. Wang et al. [8] analyzed the influence of control gain parameters on system stability under random excitation induced by track irregularities, determining recommended value ranges for proportional derivative (PD) control parameters.

Beyond classical control methods like full-state feedback and PD control, modern control techniques have been widely applied to enhance levitation system robustness. For example, considering voltage saturation, sensor noise, and various parameter uncertainties in the levitation system, Li et al. [9] developed a self-tuning double-layer SMC system capable of achieving high-performance control with relatively low control voltages under significant model uncertainty and disturbances, validated experimentally. Considering constraints on gap, velocity, and acceleration, Hu et al. [10] proposed a robust MPC strategy, efficiently solving the optimization problem using a primal-dual interior-point algorithm to handle constraints in real-time, with effectiveness demonstrated via simulation. Furthermore, incorporating vehicle-track coupled vibration and control loop faults, Guo et al. [11] established a coupled vehicle-track model, analyzed the dynamic response performance of the levitation system under different fault conditions, and validated the model's accuracy by comparing it with measured train-viaduct responses.

State observers, which reconstruct unmeasurable system states from measurable outputs, provide more complete state information for controller design [12]. The application of sliding mode observers [13], extended state observers [14], and Kalman filter observers [15] has further improved the control accuracy and disturbance rejection capability of levitation systems under complex operating conditions. For instance, considering track irregularity excitation, Jiang et al. [16] designed an adaptive disturbance observer to compensate for and suppress irregularity disturbances, achieving stable levitation of a half-suspension frame

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combined with a prescribed-time synchronous controller. Zhu et al. [17] proposed an adaptive SMC method based on a linear extended state observer (LESO). The LESO estimates the system's total disturbance and unmodeled dynamics in real-time, which are then canceled within the SMC. Simulations and experiments verified the controller's strong robustness against noise and external disturbances.

Unlike the model-based methods above, learning-based methods can adaptively learn optimal control policies directly from data, freeing them from reliance on manual tuning and expert experience, thus offering better adaptability. Representative learning-based methods include iterative learning control [18], neural network control [19], and reinforcement learning control (RL) [20]. Particularly, deep RL, with its capability to learn complex policies in high-dimensional environments, has achieved remarkable success and become a powerful approach for addressing challenges in levitation systems. For example, Zhu et al. [21] proposed a multi-agent RL cooperative levitation controller, with simulations indicating its effectiveness in handling coupling effects between levitation units. While these methods yield promising results, the following issues remain unresolved:

1) In model-based methods, controller design heavily depends on accurate mathematical models of the levitation system, and performance is significantly influenced by manual tuning and expert experience. Adapting hyperparameters to accommodate the changing dynamics of the levitation system in complex environments is challenging.

2) Learning-based methods typically employ an end-to-end approach to directly solve for the optimal policy. However, under strong disturbances like aerodynamic loads, RL algorithms struggle to converge and find the optimal solution.

3) Current RL algorithms designed for levitation systems lack safety guarantees, unable to prevent the agent from taking unsafe actions during random exploration.

Notably, the concept of safe RL has been explored in other domains. For instance, Emam et al. [22] formulated a constrained optimization problem based on control barrier functions (CBF) to ensure safety during training. Han et al. [23] constructed cost constraints based on Lyapunov theory, theoretically guaranteeing the stability of the control system upon RL convergence. Synthesizing the above analysis, this paper focuses on the disturbance caused by aerodynamic lift on high-speed maglev train levitation systems and proposes a safe RL control method based on higher-order CBF (HOCBF) and disturbance observer (DOB). Guided by a hierarchical design principle, actions generated by the RL agent are first corrected by a safety layer constructed using HOCBF. Additionally, a DOB is designed to compensate for aerodynamic loads, ensuring stable algorithm convergence. The main contributions of this paper are:

1) The proposed method adaptively learns the optimal policy from data, avoiding the high model dependency of existing approaches.

2) The safety layer, designed using HOCBF, theoretically guarantees the safety of the levitation system. Furthermore, embedding the safety layer into the RL algorithm via differentiable optimization accelerates convergence.

3) Targeted handling of disturbances like aerodynamic loads through the DOB mitigates convergence difficulties of the RL algorithm caused by strong disturbances, improving algorithm stability and control performance.

The rest of this paper is organized as follows: Section II establishes the single-point levitation system model and formulates it as a Markov Decision Process (MDP). Section III introduces the adopted RL algorithm, provides a detailed derivation of the proposed HOCBF-based safe RL control algorithm for the levitation system, and presents the observer design alongside stability proofs. Section IV validates the effectiveness of the proposed method under three different operating conditions, including aerodynamic load disturbances, via simulation. Section V concludes the paper.

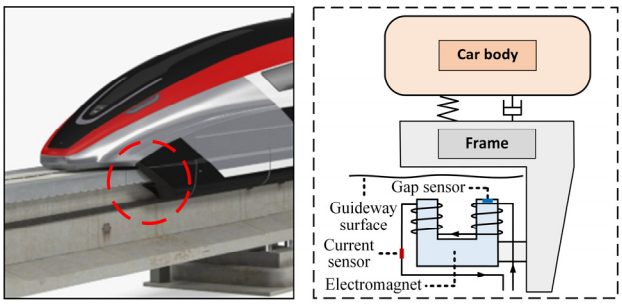


Fig. 1. Levitation system model

II. Levitation System Modeling

Considering the levitation system model shown in Fig. 1, the electromagnetic force can be expressed as

$$F_e = \frac{\mu_0 N^2 S}{4}\left(\frac{u(t)}{c(t)}\right)^2 \tag{1}$$

where μ_0 denotes the magnetic permeability of air, N represents the number of turns of the electromagnet coil, S denotes the pole area, $u = i$ refers to the control current, and c denotes the levitation air gap.

The force transmitted from the suspension system to the electromagnet can be expressed as

$$F_s = k_s(z_f - z_c) + k_c(\dot{z}_f - \dot{z}_c) \tag{2}$$

where k_s is the suspension stiffness, k_c is the suspension damping coefficient, z_f is the car body displacement, z_c is the electromagnet displacement, $z_c = c$.

Selecting the state vector as $z = [z_1 \quad z_2 \quad z_3 \quad z_4] = [c \quad \dot{c} \quad z_f \quad \dot{z}_f]$, the system equation can be written as

$$\begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = -\dfrac{F_e}{m} + g + \dfrac{F_s}{m} \\ \dot{z}_3 = z_4 \\ \dot{z}_4 = g - \dfrac{F_s}{M} + \dfrac{F_d}{M} \end{cases} \tag{3}$$

where m denotes the mass of the electromagnet, M denotes the equivalent mass of the car body mapped to the

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single-point levitation system, and F_d represents the disturbance force.

Defining the levitation controller as the agent and the levitation system as the environment, the problem is modeled using a MDP. The MDP is represented by the tuple $\langle \mathcal{Z} \quad \mathcal{U} \quad P \quad R \rangle$, consisting of: a set $\mathcal{Z}$ composed of state variables z; a set $\mathcal{U}$ composed of actions $u = i$; the state transition probability $P(z(t+1) \mid z(t), u(t))$ defining the probability of transitioning from the current state to the next state; and the reward function $R = r(z,u)$. At time step t, the agent generates a control input $u(t) = i(t)$ via its policy $\pi_\theta(u(t) \mid z(t))$ and interacts with the environment, causing the system state to transition from $z(t)$ to $z(t+1)$ and receiving a reward $R(t)$. The tuple $\big(z(t) \quad u(t) \quad R(t) \quad z(t+1)\big)$ obtained from the agent-environment interaction is stored in the experience replay buffer $\mathcal{R}$.

III. CONTROL SYSTEM DESIGN

A. RL controller

The designed RL controller for the levitation system is based on the Soft Actor-Critic (SAC) algorithm. Its optimization objective can be written as:

$$\max \ \mathbb{E}_{\pi_\theta}\left[\sum_t \eta^t r(z(t),u(t))\right] \tag{4}$$

$$\text{s.t. } \mathbb{E}_{\mathcal{R}}\left[-\log(\pi(u(t) \mid z(t)))\right] \geq \mathcal{H}_0 \tag{5}$$

where π_θ is the policy; $\eta \in [0,1]$ is the discount factor; and $\mathcal{H}_0$ is an entropy threshold used to constrain the stochasticity of the policy.

For continuous action spaces, such as the levitation system, SAC outputs the mean and variance of a Gaussian distribution. Differentiability relative to the policy network is achieved using the reparameterization trick:

$$u(t) = f_\theta(\epsilon(t), z(t)) \tag{6}$$

where ϵ is a noise variable sampled from a unit Gaussian distribution $\mathcal{N}$, and f_θ represents the policy network.

The loss function for the policy network is:

$$\begin{aligned} &L_\pi(\theta) \\ &= \mathbb{E}_{\mathcal{R}}[\alpha \log(\pi_\theta(u(t) \mid z(t))) - \min_{j=1,2} Q_{\omega_j}(z(t),u(t))] \end{aligned} \tag{7}$$

where α is the entropy weight coefficient, and Q_ω is the action-value function (Q-function). To mitigate the overestimation of action values, two Q-function networks are used, and the minimum of their outputs is taken.

The loss function for any one Q-function network is:

$$\mathbb{E}_{\mathcal{R}}[\frac{1}{2}(Q_\omega(z(t),u(t)) - (R(t) + \eta V_{\bar{\omega}}(z(t+1))))^2] \tag{8}$$

where:

$$\begin{aligned} &V_{\bar{\omega}}(z(t+1)) \\ &= \min_{j=1,2} Q_{\bar{\omega}_j}(z(t+1),u(t+1)) \\ &-\alpha \log \pi_\theta(u(t+1) \mid z(t+1)) \end{aligned} \tag{9}$$

The loss function for the entropy weight coefficient is:

$$L(\alpha) = \mathbb{E}\left[-\alpha \log \pi_\theta(u(t) \mid z(t)) - \alpha \mathcal{H}_0\right] \tag{10}$$

B. Safety layer based on HOCBF

Directly applying the above RL algorithm to the levitation system will inevitably lead to collisions accidents between the electromagnet and the track during the training phase due to random exploratory actions. Therefore, a safety layer is designed to modify the actions generated by the RL agent, ensuring the system always remains within a safe state.

Define the safe set $\mathcal{S} \subset \mathbb{R}^n$ as:

$$\begin{cases} \mathcal{S} = \{z \mid h(z) \geq 0\} \\ \partial\mathcal{S} = \{z \mid h(z) = 0\} \\ \text{Int}(\mathcal{S}) = \{z \mid h(z) > 0\} \end{cases} \tag{11}$$

where $h : \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable function used to determine if the system is within the safe set; $\partial\mathcal{S}$ represents the boundary of the safe set, and $\text{Int}(\mathcal{S})$ represents the interior of the safe set.

For the levitation system, $h(z)$ can be defined as:

$$h(z) = \frac{1}{2}(z_d^2 - z_1^2) \tag{12}$$

where z_d is the target levitation gap.

To construct the safety layer, consider the suspension system and the electromagnet in system equation (3) as an integrated whole. Linearize the system near the equilibrium point and express it in affine form:

$$\dot{z} = f(z) + g(z)u + d \tag{13}$$

where $z = \begin{bmatrix} z_1 & z_2 \end{bmatrix} = \begin{bmatrix} \Delta c & \Delta \dot{c} \end{bmatrix}$; $u = \Delta i$; d represents disturbances; $f(z) = \begin{bmatrix} z_1 & \dfrac{K_c}{m+M} z_2 \end{bmatrix}^{\mathrm{T}}$;

$g(z) = \begin{bmatrix} 0 & -\dfrac{K_i}{m+M} \end{bmatrix}^{\mathrm{T}}$; $K_c = \left.\dfrac{\partial F_e}{\partial c}\right|_{(i_0,c_0)} = \dfrac{\mu_0 N^2 S i_0^2}{2c_0^3}$;

$K_i = \left.\dfrac{\partial F_e}{\partial i}\right|_{(i_0,c_0)} = \dfrac{\mu_0 N^2 S i_0}{2c_0^2}$; c_0 is the equilibrium gap; and i_0 is the equilibrium current.

A set is called forward invariant if the system state, starting from within the safe set, remains within the set for all future time. The following definition uses $h(z)$ to construct constraints that guarantee system safety.

Definition 1: Consider the safety function $h : \mathbb{R}^n \to \mathbb{R}$, with relative degree $\gamma \in \mathbb{Z}_+$. Recursively define a sequence of functions $\varphi_i : \mathbb{R}^n \to \mathbb{R}(i = 0,1,\ldots,\gamma)$:

$$\begin{cases} \varphi_0(z) = h(z) \\ \varphi_i(z) = \dot{\varphi}_{i-1}(z) + \beta_i(\varphi_{i-1}(z)), \quad i = 1,\ldots,\gamma \end{cases} \tag{14}$$

where $\beta_i : \mathbb{R} \to \mathbb{R}(i = 1,\ldots,\gamma)$ are class $\mathcal{K}$ functions, and β_i is $(\gamma - i)$ times differentiable (β_γ only needs to be a class $\mathcal{K}$ function).

The corresponding sequence of safety sets is defined as:

$$C_i = \{\varphi_{i-1}(z) \geq 0\}, \quad i = 1,\ldots,\gamma \tag{15}$$

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If there exist class $\mathcal{K}$ functions $\{\beta_i\}_{i=1}^{\gamma}$ satisfying differentiability requirements, such that for all $z \in \bigcap_{i=1}^{\gamma} C_i$, there exists a control input $u \in \mathcal{U}$ satisfying:

$$L_f^{\gamma} h(z) + L_g L_f^{\gamma-1} h(z) u + D(h(z)) \geq 0 \tag{16}$$

where $D(h(z)) = \sum_{i=1}^{\gamma-1} L_f^i \left(\beta_{\gamma-i} \circ \varphi_{\gamma-i-1}\right)(z) + \beta_{\gamma}(\varphi_{\gamma-1}(z))$.

Then, $h(z)$ is called a γ order HOCBF for the system. This guarantees the forward invariance of the system, meaning it always remains within the safe set.

From (13), the system's relative degree is $\gamma = 2$. Then, from (14), we get:

$$\begin{cases} \varphi_0(z) = h(z) \\ \varphi_1(z) = \dot{h}(z) + \beta_1(h(z)) \end{cases} \tag{17}$$

Substituting (17) into (16) yields:

$$\begin{aligned} &L_f^2 h(z) + L_g L_f h(z) u + L_f \beta_1(h(z)) \\ &+ \beta_2(\dot{h}(z) + \beta_1(h(z))) \geq 0 \end{aligned} \tag{18}$$

Substituting (13) and (12) into (18) gives the effective HOCBF condition for the levitation system:

$$\begin{aligned} &-\frac{K_c}{m+M} z_1^{\,2} - z_2^{\,2} + \frac{K_i}{m+M} z_1 u \\ &-2\lambda \cdot x_1 x_2 + \lambda^2 \cdot \frac{1}{2}(x_d^{\,2} - x_1^{\,2}) \geq 0 \end{aligned} \tag{19}$$

where 2λ and λ^2 serve as class $\mathcal{K}$ functions.

Integrating with the SAC algorithm described above, the modification of the RL action by the safety layer can be described as the following constrained optimization problem, solvable via Quadratic Programming (QP):

$$\begin{aligned} &(u_{\mathrm{CBF}}^*(u_{\mathrm{RL}}(z), z), \xi) \\ &= \arg\min \left\| u_{\mathrm{CBF}}(u_{\mathrm{RL}}(z), z) \right\| + l\xi^2 \end{aligned} \tag{20}$$

$$\begin{aligned} \text{s.t. } &-\frac{K_c}{m+M} z_1^{\,2} - z_2^{\,2} + \frac{K_i}{m+M} z_1 (u_{\mathrm{CBF}} + u_{\mathrm{RL}}) \\ &\geq 2\lambda \cdot z_1 z_2 - \lambda^2 \cdot \frac{1}{2}(z_d^{\,2} - z_1^{\,2}) + \xi \end{aligned} \tag{21}$$

where u_{CBF}^* denotes the correction quantity solved through the CBF, u_{RL} represents the control input given by the RL agent, $l > 0$ is the weighting coefficient, and $\xi \in \mathbb{R}$ is the slack variable used to ensure the feasibility of QP.

After embedding the safety layer, the loss function for the SAC policy network is modified to:

$$\begin{aligned} L_{\pi}(\theta) = \mathbb{E}_{\mathcal{R}}[&\alpha \log(\pi_{\theta}(u_{\mathrm{RL}}(t) \mid z(t))) \\ &- \min_{j=1,2} Q_{\omega_j}(z(t), u_{\mathrm{RL}}(t) + u_{\mathrm{CBF}}^*(t))] \end{aligned} \tag{22}$$

The loss function for any one Q-function network becomes:

$$\begin{aligned} &\mathbb{E}_{\mathcal{R}}[\frac{1}{2}(Q_{\omega}(z(t), u_{\mathrm{RL}}(t) + u_{\mathrm{CBF}}^*(t)) \\ &-(R(t) + \eta V_{\bar{\omega}}(z(t+1))))^2] \end{aligned} \tag{23}$$

where

$$\begin{aligned} &V_{\bar{\omega}}(z(t+1)) \\ &= \min_{j=1,2} Q_{\bar{\omega}_j}(z(t+1), u_{\mathrm{RL}}(t+1) + u_{\mathrm{CBF}}^*(t+1)) \\ &-\alpha \log \pi_{\theta}(u_{\mathrm{RL}}(t+1) \mid z(t+1)) \end{aligned} \tag{24}$$

The control input finally applied to the levitation system is then:

$$u^* = u_{\mathrm{CBF}}^* + u_{\mathrm{RL}} \tag{25}$$

C. DOB design

The disturbance in (13) is handled using a DOB. Let $d_T = -d$, then the observer equations are:

$$\dot{\hat{d}}_T = \xi_1(\hat{z}_2 - z_2) \tag{26}$$

$$\dot{\hat{z}}_2 = -\hat{d}_T + \frac{K_c}{m+M} z_2 - \frac{K_i}{m+M} u - \xi_2(\hat{z}_2 - z_2) \tag{27}$$

where $\hat{d}_T$ is the estimate of the disturbance d_T, $\hat{z}_2$ is the estimate of the velocity z_2, and $\xi_1, \xi_2 > 0$ is the gain coefficient.

The algorithm is shown in Fig. 2.

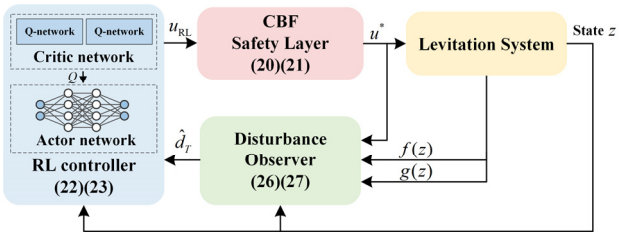


Fig. 2. Block diagram of the algorithm.

Stability Analysis: Define the observation errors as:

$$\begin{cases} \tilde{d}_T = d_T - \hat{d}_T \\ \tilde{z}_2 = z_2 - \hat{z}_2 \end{cases} \tag{28}$$

Define the following Lyapunov function:

$$V_{\mathrm{L}} = \frac{1}{2\xi_1} \tilde{d}_T^2 + \frac{1}{2} \tilde{z}_2^{\,2} \tag{29}$$

Taking its derivative yields:

$$\begin{aligned} \dot{V}_{\mathrm{L}} &= \frac{1}{\xi_1} \tilde{d}_T \dot{\tilde{d}} + \tilde{z}_2 \dot{\tilde{z}}_2 \\ &= \frac{1}{\xi_1} \tilde{d}_T (\dot{d}_T - \dot{\hat{d}}_T) + \tilde{z}_2 (\dot{z}_2 - \dot{\hat{z}}_2) \end{aligned} \tag{30}$$

Considering the aerodynamic load as a slow time-varying signal and noting that the observer gain is generally chosen large, (30) can be rewritten as:

$$\dot{V}_{\mathrm{L}} = -\frac{1}{\xi_1} \tilde{d}_T \dot{\hat{d}}_T + \tilde{z}_2 (\dot{z}_2 - \dot{\hat{z}}_2) \tag{31}$$

Substituting (26), (27), and (28) into (31) gives:

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$$\begin{aligned}\dot{V}_L &= -\frac{1}{\xi_1}\tilde{d}_T(\xi_1(\hat{z}_2 - z_2)) \\ &+\tilde{z}_2(-d_T + \frac{K_c}{m+M}z_2 - \frac{K_i}{m+M}u \\ &+\hat{d}_T - \frac{K_c}{m+M}z_2 + \frac{K_i}{m+M}u + \xi_2(\hat{z}_2 - z_2)) \\ &= \tilde{d}_T\tilde{z}_2 + \tilde{z}_2(\hat{d}_T - d_T - \xi_2(z_2 - \hat{z}_2)) \\ &= -\xi_2\tilde{z}_2^{\,2}\end{aligned} \tag{32}$$

Since $\xi_2 > 0$, then $\dot{V} \le 0$. Thus, the observer is stable.

IV. Simulation Analysis

The effectiveness of the proposed method was validated through numerical simulations based on the parameters of the single-point levitation system model. The model parameters are listed in TABLE I.

The RL agent training algorithm was implemented in Python. Both the simulation step and control step were set to 0.001 seconds. The maximum number of training steps per episode was set to 1000. Evaluation was performed every 5 episodes, and agents achieving episode rewards greater than 950 were saved. Both the Actor and Critic networks were designed with two hidden layers, each containing 256 neurons. The mini-batch size was 256. The reward function is designed as

$$r(z) = 0.8^{(1000\cdot z_1)^2} + 0.2\cdot 0.8^{z_2^2} \tag{33}$$

The reward convergence curve during training is shown in Fig. 3.

TABLE I
Model Parameters

Parameter	Value
Mass of car body	1500 kg
Mass of electromagnet	750 kg
Coil turns	270
Coil area	0.115 m²
Target of gap	10 mm
Vacuum permeability	$4\pi\times10^{-7}$ H/m

Simulation analyses were conducted under three different operating conditions: 1) static levitation without disturbances; 2) aerodynamic load disturbance; 3) combined aerodynamic load and track irregularity disturbance. The proposed method is respectively compared with a Proportional-Integral-Derivative (PID) controller and PID controller with DOB (PID-DOB). The PID parameters are set as $k_p = 55$, $k_i = 5$, $k_p = 1.5$.

Track irregularity was simulated using a sine wave. The aerodynamic load was modeled as:

$$F_a = \frac{1}{2}\rho A C_z v^2 \tag{34}$$

where ρ is the air density, $A = 12\text{m}^3$ is the frontal area, $v = 600$km/h is the relative wind speed, and C_z is the aerodynamic force coefficient, defined as:

$$\begin{aligned}C_z &= -4.54\times10^{-6}v_w^3 + 7.32\times10^{-4}v_w^2 \\ &+7.55\times10^{-3}v_w + 0.7364\end{aligned} \tag{35}$$

where $v_w = 20$m/s is the crosswind speed.

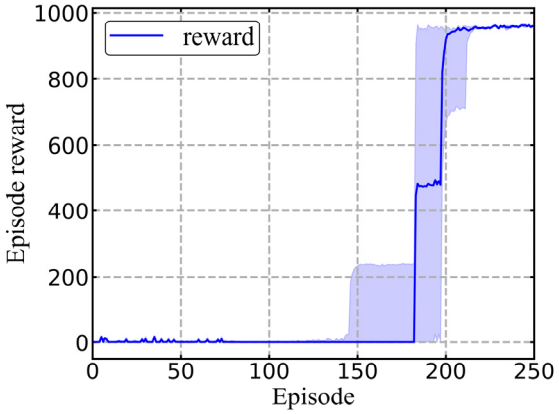


Fig. 3. Reward convergence curve

A. Static Levitation

Fig. 4 illustrates the response of the levitation air gap. In the absence of any disturbances, all three methods achieved satisfactory control performance. The responses of PID and PID-DOB were similar, completing static levitation in approximately 0.19 s. The proposed method responded faster, with a settling time of about 0.04 s, but exhibited a slight steady-state error. The electromagnet current response is shown in Fig. 5. The responses of the PID and PID-DOB methods were smoother, while the proposed method showed some oscillation. Overall, under disturbance-free conditions, all three methods delivered good control performance.

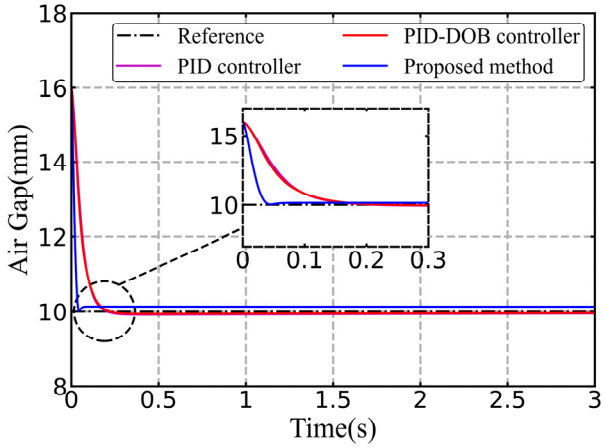


Fig. 4. Case1: air gap response

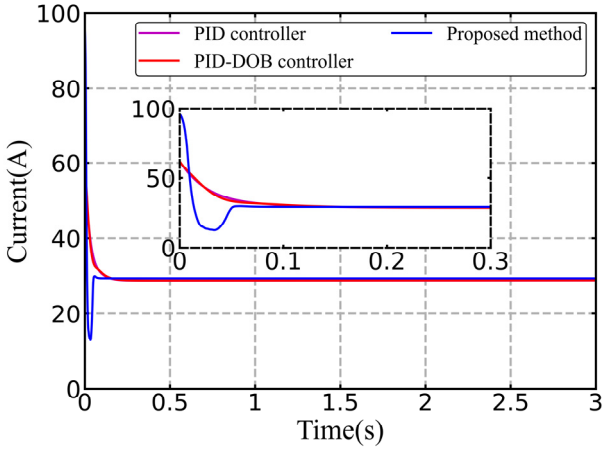


Fig. 5. Case1: electromagnet current response

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B. *Aerodynamic load disturbance*

The levitation air gap response under aerodynamic lift force disturbance is shown in Fig. 6. The PID-controlled gap was significantly affected by the aerodynamic lift disturbance, rapidly decreasing away from the equilibrium point, with a maximum error of 3.26 mm and an RMSE of 2.37 mm. The PID-DOB method was noticeably less affected than PID control, achieving a maximum error of 0.83 mm and an RMSE of approximately 0.10 mm. It adjusted rapidly after deviation and stabilized at the equilibrium point. In comparison, the proposed method demonstrated superior resistance to wind load disturbance, with a maximum error of only 0.34 mm and an RMSE of only 0.06 mm. This represents reductions of 58.4% and 42.6%, respectively, compared to PID-DOB.

The electromagnet current response is shown in Fig. 7. The proposed method and PID-DOB exhibited similar responses, while the PID method deviated from the equilibrium point due to the disturbance and adjusted slowly. The acceleration response is shown in Fig. 8, where the proposed method exhibited relatively smaller acceleration fluctuations. Overall, under aerodynamic load disturbance, the PID method could not effectively reject the disturbance, causing both the levitation air gap and electromagnet current to deviate from the equilibrium point. Both the PID-DOB method and the proposed method demonstrated good disturbance rejection capabilities, with the proposed method achieving further performance improvements over PID-DOB.

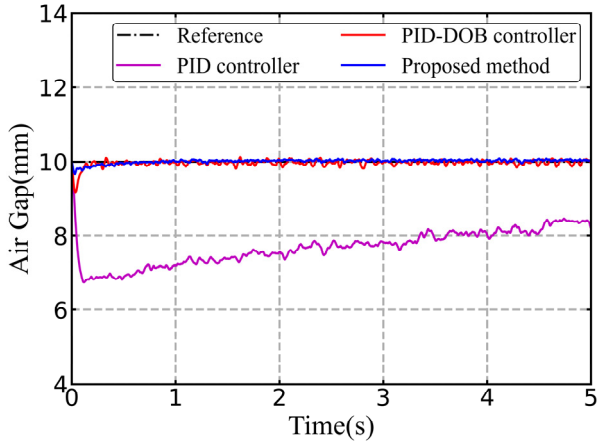


Fig. 6. Case2: air gap response

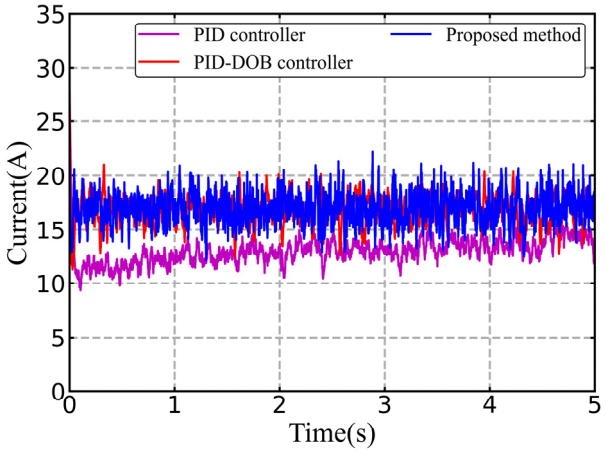


Fig. 7. Case2: electromagnet current response

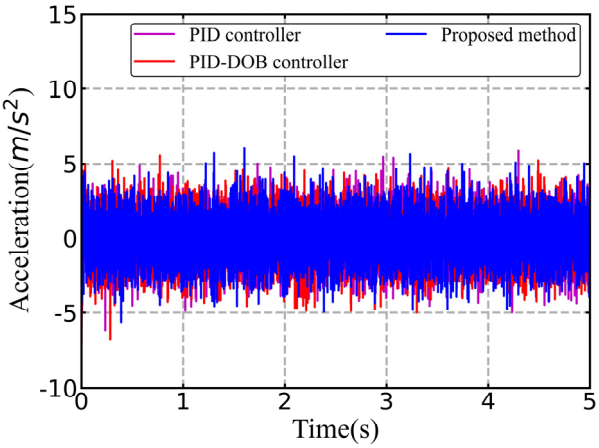


Fig. 8. Case2: acceleration response

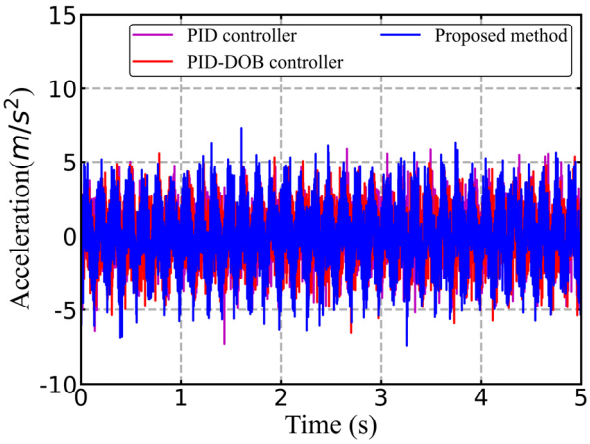


Fig. 11. Case3: acceleration response

C. *Combined aerodynamic load and track irregularity disturbance*

The levitation air gap response under aerodynamic lift force and track irregularity disturbances is shown in Fig. 9. PID control was the most affected by the disturbances, deviating from the equilibrium point and exhibiting large gap fluctuations. Both the PID-DOB method and the proposed method stabilized around the equilibrium point. Comparatively, the proposed method was less affected by the irregularities, achieving a maximum error of 0.93 mm and an RMSE of 0.40 mm. This represents reductions of 27.3% and 46.1%, respectively, compared to PID-DOB. The electromagnet current response is shown in Fig. 10. The current fluctuation of the proposed method was larger. Similarly, as shown in the acceleration response (Fig. 11), the fluctuation of the proposed method was slightly larger than the other methods.

The performance metrics of the PID, PID-DOB, and proposed methods under the three cases are summarized in TABLE II. It is evident that the proposed method outperforms the others across various operating conditions, demonstrating superior capability in rejecting aerodynamic loads and track irregularities.

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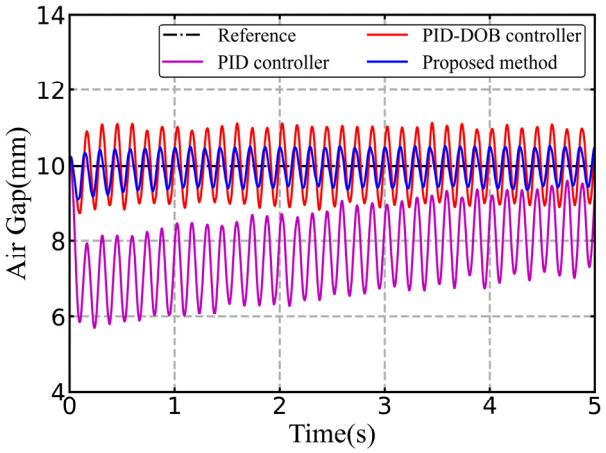


Fig. 9. Case3: air gap response

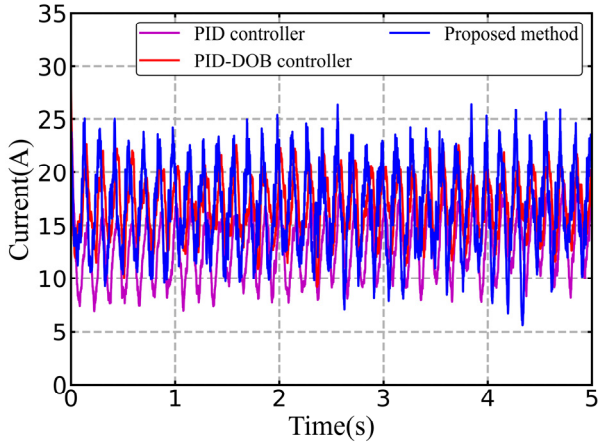


Fig. 10. Case3: electromagnet current response

TABLE II
SYSTEM PERFORMANCE METRICS

		PID	PID-DOB	Proposed method
Case1	RMSE (mm)	0.6456	0.6284	0.3975
	Adjustment Time (s)	0.19	0.19	0.04
Case2	RMSE (mm)	2.3686	0.0996	0.0572
	Max error (mm)	3.2617	0.8270	0.3442
Case3	RMSE (mm)	2.5053	0.7400	0.3986
	Max error (mm)	4.3067	1.2758	0.9276

V. CONCLUSION

Addressing the stability and safety challenges faced by high-speed maglev train levitation systems under complex disturbances such as aerodynamic lift forces and track irregularities, this paper proposed a safe RL control method integrating HOCBF and DOB. The method employs a hierarchical design: the RL agent adaptively learns the optimal control policy from data, reducing reliance on precise mathematical models; the safety layer constructed using HOCBF rigorously guarantees system safety by modifying the agent's actions in real-time via differentiable optimization; and the DOB specifically compensates for external disturbances like aerodynamic loads, significantly enhancing the convergence stability and control accuracy of the algorithm under strong disturbances. Simulation results demonstrate that under three typical operating conditions: static levitation, aerodynamic load disturbance alone, and combined aerodynamic load with track irregularity, the proposed method significantly outperforms traditional PID and PID-DOB in key performance metrics such as RMSE and maximum error of the levitation air gap. The advantage is particularly prominent in rejecting aerodynamic load disturbances. This method effectively resolves the issues inherent in existing model-based approaches and end-to-end RL methods, providing a reliable control strategy for the safe and stable operation of high-speed maglev trains. Future work could involve Hardware-in-the-Loop testing to further refine the control strategy.

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